

# Image Warping and Mosaicing - Part A

## Shoot images and recover homographies

I used these images throughout my project:



Ihouse Great Hall 1



Ihouse Great Hall 2



Ihouse Elevator 1



Ihouse Elevator 2



## Ihouse Stairs 1

## Ihouse Stairs 2

The first step of the project was to define correspondances between the images. To do this, I used the tool from project 3. I observed that for best results, it was preferable to have a large number of point pairs and have a significant overlap between images.

Once I have defined the correspondances, the next task is to recover a transformation such that:

$$\begin{bmatrix} a & b & c \\ d & e & f \\ g & h & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} = \begin{bmatrix} wx' \\ wy' \\ w \end{bmatrix}$$

Note that this transformation won't map points from one image to another directly, but there will also be a scaling component  $w$  which we had to account for (it took me some time to realize this). From here, we expand and rearrange to obtain:

$$\begin{bmatrix} x & y & 1 & 0 & 0 & 0 & -xx' & -yx' \\ 0 & 0 & 0 & x & y & 1 & -xy' & -yy' \end{bmatrix} \begin{bmatrix} a \\ b \\ c \\ d \\ e \\ f \\ g \\ h \end{bmatrix} = \begin{bmatrix} x' \\ y' \end{bmatrix}$$

Note that if we pick four points there will be an exact solution to this system. However, in practice, due to inaccuracy in point selection it is better to select more points and then use least squares.

## Warp images

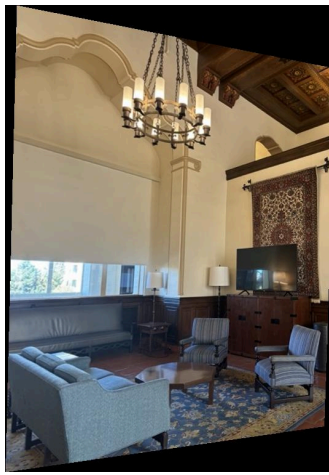
Once we have recovered the homography  $H$ , we can use this matrix to warp one of the image (which is necessary before mosaicing). To do this, I reused some of the techniques from project 3. Specifically, the steps to warp an image are the following:

- First, we apply the homography to the corners of the image to establish the shape of the final warp.
- Then, using `sk.draw.polygon`, we find the coordinates of the pixels in this resulting shape. We can perform an inverse warp of these coordinates to recover the corresponding positions in the source image.
- Finally, we interpolate the pixel values at these positions and fill in the result.

For example, here is the first great hall image warped according to its homography with great hall 2:



Ihouse great hall 1  
Original



Ihouse great hall 1  
warped

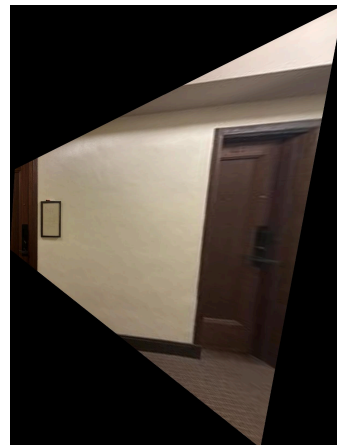


Ihouse Great Hall 2

We can get interesting results by warping images that contain rectangular objects in perspective into images that force those objects to appear rectangular. To do this, we simply define a correspondance between the four corners and a rectangle of our choice:



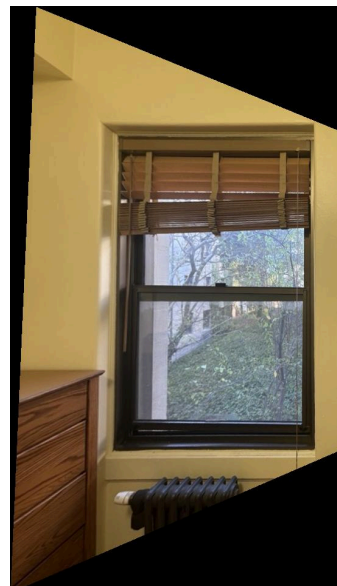
Board Outside my  
Room



Warped Board



Book on chest



## Warped Window

Note that it might seem that the shapes are still not rectangular. I believe that this is a visual effect due to the distorted perspective.

## Mosaicing

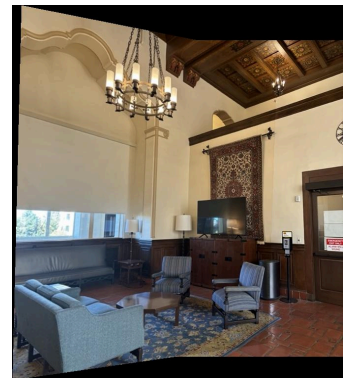
Once we have recovered the homography matrix, we may blend one image into another by superposing the aligned homography with the original destination image. To do this, we start calculating the size of the final projection which is calculated from the sizes of the images and the offset between them. After building the 'canvas' where the result will go, we may superimpose the images. To avoid the appearance of edges between the images, we blend them using an alpha mask that decreases from 0 to 1 between images. We obtain the following results:



Leftmost image



Rightmost image



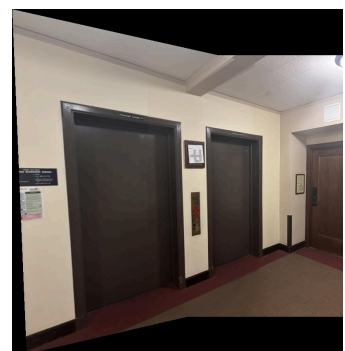
Mosaic



Leftmost Image



Rightmost image



Mosaic





Leftmost Image



Rightmost image



Mosaic

Note that there is no additional difficulty to doing 3 images instead of 2. We obtain the homography between 1 and 3 by multiplying those of 1 and 2 and 2 and 3. I made a naive (not alpha blended) 3 photo mosaic:



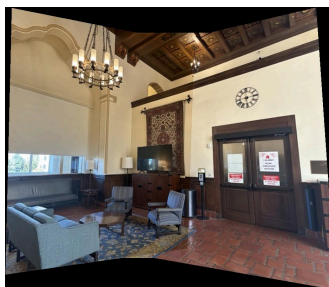
Leftmost image



Middle image



Rightmost image



Naive mosaic